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SETI and Post-Detection: Towards a New Research Roadmap

Kathryn Denning^{a*}, Anamaria Berea^b, Klara Anna Capova^c, Duncan Forgan^d,
Eric Korpela^e, Michael P. Oman-Reagan^f, Jill Tarter^g, Leslie I. Tennen^h

^aYork University, Canada, kdenning@yorku.ca

^bBlue Marble Space Institute of Science, USA, anamaria@bmsis.org

^cUniversity of Durham, UK, k.a.capova@durham.ac.uk

^dEdinburgh, UK, dh4gan@gmail.com

^eBerkeley SETI Research Center, UC Berkeley, USA, korpela@berkeley.edu

^fMemorial University, Canada, omanreagan@mun.ca

^gSETI Institute, USA, tarter@seti.org

^hLaw Offices of Sterns and Tennen, USA, ltennen@astrolaw.com

*Corresponding Author

Abstract

Since SETI began, there has been consistent multidisciplinary interest in “post-detection” matters. A considerable literature exists, spanning decades and many domains of research. However, profound changes in and around SETI/technosignatures work, recent efforts at revising the Rio Scale, and proposals to revise the Protocols (i.e., “Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence”) have sparked a renewed multidisciplinary conversation. There is an urgent need for effective interdisciplinary collaboration which brings SETI scientists, social scientists, humanists, legal experts, science policymakers, risk management experts, et al., into a sustained and sustainable conversation about SETI in society as the world changes. This multi-authored, multi-disciplinary discussion paper (not necessarily consensus) is an effort in that direction, bringing social scientists, a legal expert, and SETI/technosignatures researchers together in a conversation to ask: what basic research needs to be done, now, to fill in our knowledge and guide policy about crucial post-detection matters?

Here, we rethink the catch-all notion of “post-detection” as three entangled themes: preparation for discovery, process of discovery, and protocol during detection/discovery. We emphasize that discovery is best understood as a process, not a moment, and that different expertises are applicable throughout. We briefly review prior and recent work on “post-detection” themes, and suggest some general directions for supplementing the Protocols with Codes of Conduct and Best Practices guidelines. Aspects of the current global context for SETI/technosignatures research are briefly covered, including concepts of science and social responsibility, challenges for trust, and the increasingly complex existential risk landscape. We sketch areas for research roadmapping, including: social media safety for researchers; the increasing diversity of active players in the research (e.g. professionals, students, citizen scientists); habits of thought in the SETI/technosignatures community; prevalent ideas about alien characteristics and behavior; widespread popular notions about detection and its sequelae; and previous examples of claimed detections. We explore several areas in more detail, including: public surveys concerning extraterrestrials and contact; social media and information transmission; applicable lessons from mass warnings/risk communication systems; economic impacts; and legal issues. Finally, we suggest research priorities focusing upon best practices guidelines in areas of the most urgent concern.

Disclaimer: This paper represents the collaborative efforts of the authors in our individual capacities to consider the subject from a variety of viewpoints, and does not necessarily reflect a consensus on all points raised. Furthermore, the views contained herein are not to be considered as expressing the position of any organizations or institutions with which we may be associated or employed.

Keywords: technosignatures, SETI, post-detection, societal impact, interdisciplinary, astrobiology

1. Introduction

This paper takes a step back to reconsider the challenges of what has historically been called “post-detection” in the SETI/technosignatures community, i.e. the sequelae to a detection of technological

extraterrestrial life. We started from the widely supported assumption [1]¹ that a wise approach involves forethought, and asked ourselves: what basic research needs to be done, now, to fill in our knowledge, and guide policy and practice about crucial

¹ Denning and Dick 2019 [1] received rapid and widespread endorsement from the SETI community, confirming that this is still a widely held assumption today, as it was earlier in SETI’s history.

post-detection matters? What kinds of expertise can be brought to bear on the challenges instigated by an astronomy research program of wide public interest and potentially high impact? Below, we outline the results of our collaborative thought experiment. We do not prescribe a set of practices, protocols, or rules for post-detection scenarios. Rather, we ask what the known key challenges are, what research could better define those challenges, and what research could determine how those challenges would best be addressed. This paper thus sketches out some preliminaries for a research roadmap; in conjunction with prior efforts, it could serve as a baseline for full roadmapping exercises in the future. In turn, the research that such a roadmap would foster, could help with generating practical documents such as codes of conduct, best practices guidelines for social media use and information dissemination, and updated SETI principles/protocols. Such research cannot be done just once, because both SETI/technosignatures searches [2,3,4] and the world are changing continually and quickly. Iteration is key. Meanwhile, recognizing that a full, multi-expert research roadmapping process may take over a year (and is contingent upon workshop funding), we have suggested some priority areas which should be tackled as soon as possible.

To begin our step back: Astronomers recognize the need for different approaches to the phases of detection. Experts from many other disciplines are well-suited to work alongside astronomers in this effort and help inform policy discussions relevant to all phases of detection, from preparation to protocols. The big questions in SETI about intelligence, technology, civilization, evolution, etc. are both astronomy topics and social topics, a fact SETI scientists have long acknowledged through their regular collaborations with social scientists [5,6]. The phases of the detection process are, likewise, both an astronomy problem and a social problem and they require the expertise of both the SETI community and experts from other disciplines and practices who specialize in studying the social.

The astronomy community increasingly understands a key point made by SETI historian Steve Dick [7], Cirkovic [8] and others: that discovery isn't a moment, but a process. Accordingly, we need to reframe "post-detection" for at least two reasons: detection is not one simple temporal moment with a before and after; and much of the relevant work must happen all the time, starting now, not just "someday". Detection presents multiple challenges around distribution of information [9] and the actions required during the discovery process itself. Although in one sense, detection would unfold in linear time, each challenge during detection would surface at multiple points along the timeline. Detection is more of a

network problem. We could think of it in terms of three themes which are entangled but different: preparation + process of detection + protocol. We expand below.

The present team of authors includes representation from the disciplines and professions of astronomy, law, anthropology, economics, and data science, and blends insights from different generations, nationalities, and genders. Our length of involvement in SETI/astrobiology-related work ranges from several decades to several years. We are, in alphabetical order:

- an economist and data scientist specialising in complex systems and communication (Berea)
- an early-career anthropologist focusing on the sociocultural aspects of SETI, astrobiology, space exploration, and science policy (Capova)
- a mid-career anthropologist/archaeologist focusing on societal issues in SETI and astrobiology, scientists' uses of history, and public mis/understanding of science (Denning)
- an early-career astronomer/astrophysicist and data scientist with policy interests (Forgan)
- an early-career anthropologist who studies how interstellar exploration efforts, including SETI and astrobiology, influence human futures (Oman-Reagan)
- an astronomer working within SETI for decades with particular expertise in citizen science (SETI@home) (Korpela)
- an astronomer central in SETI research since the 1980s, with extensive international SETI policy and public outreach experience (Tarter)
- a practicing space lawyer and expert in international space law and policy, with long experience with SETI policy (Tennen)

In a full research roadmapping exercise, experts in *many* other fields should be included, particularly: psychology and neuroscience, political science, international relations, history, risk assessment and management, philosophy, ethics, science policy, science communication, education research, risk communication, sociology, public health, speculative fiction, film, games studies, visual arts and graphic design, the arts, artificial intelligence, etc.. Please see Appendix A, "How social research is done," for some background information.

In the present paper, we do not aspire to be comprehensive. We do not make specific or firm recommendations about the Protocols -- although we address some relevant components, and providing research support for such recommendations would be one obvious goal of the eventual research roadmap. Although we explore possible societal impacts of detection in the process of defining necessary research,

we minimize our specific predictions about impacts of detection, and do not address the validity of prior such discussions. We do not address METI specifically here. In this paper, we also do not address these topics, which are of pivotal importance and which we emphasize should unquestionably be included in any full roadmapping exercise:

- ethics of contact (human: extraterrestrial)
- ethics in general (human:human, and human: nonhuman animal, and human: environmental)
- studies of the evolution of intelligence and communication on Earth, both of which are essential for preparing a knowledge base with which to interpret any evidence of extraterrestrial technology or biology
- studies of language / decipherment / syntax-->meaning

The present document may be relevant to an IAA SETI Committee Post-Detection working group, if or when such a group is re/constituted – but its utility may be much broader, in that researchers beyond the IAA SETI Committee may also find it useful in honing their technosignatures-related research questions.

1.1 *Reconceptualizing post-detection:*

We propose that when thinking about preparing for detection, the term “post-detection” may not be the most useful framework now. Below, we address detection/discovery as an unfolding, ongoing flow of occurrences and interrelationships rather than a static moment or single event. Instead of one instance, detection can be considered in terms of the unfolding activity before, during, and after [7, 10]. Instead of post-detection, we suggest thinking about detection scenarios in terms of what is done in preparation for detection/discovery, the process of detection/discovery as it unfolds, and the development of protocols to use before, during, and after detection.

1. **Preparation for discovery/detection:** Preparation includes infrastructure within the SETI and broader astronomy communities such as support for researchers, communication systems, and protocols for intra-astronomy notification.
2. **Process of discovery/detection:** Detection, when considered as an emergent phenomenon, can be seen as a process instead of an event or single moment.
3. **Protocol during detection:** Protocols for during and after detection include codes of conduct, formalized best practices considering information diffusion, and detection management. These protocols should be developed in preparation, prior to the earliest stages of the detection process,

and must also be designed to incorporate new data and adjust to changing contexts and inputs.

To prepare for these phases of detection, we propose astronomers proactively consult and collaborate with experts in the issues that arise at each phase of the process. Preparation should include input from interdisciplinary experts [1, 6, 7, 11,12,13,14, 15, 20, 24, 25, 27, 31] on legal, cultural, and ethical issues, and be grounded in social studies conducted to understand the current situation and develop policy and protocols for adapting to changes in the future. Best practices concerning communication should be refined continually as media formats and possible detection types evolve. During the unfolding of detection, previously prepared infrastructures to manage information flow, media, crisis, emergency, and related issues become more active. Experts in media studies, the movement and management of information, public health, and other related areas should not only be involved in the preparation for detection but should have pre-defined decision-making roles in the teams activated as detection unfolds. Protocols during and after detection may need to be adjusted and should have feedback mechanisms built-in in order to adapt to changing contexts and developing events. While some protocols will only function if all parties follow them strictly, others may quickly become obsolete if they are not changed to adapt to changing circumstances. Legal experts, social scientists, and others who are involved in the preparation for detection should continue to be involved throughout the process of detection and throughout the use of the protocols developed during preparation. Embedding such policy and social experts within SETI organizations, institutions, and communities will enable those communities to be responsive to change and to develop new protocols as needed.

1.2 *Visually representing the research terrain*

There are many streams of research and practice which are relevant to a sequence of discovery and investigation, and societal reactions (see Fig. 1). Each stream of work has its own life, priorities, personalities, and practices: it exists continually rather than suddenly popping into existence when needed because of a discovery. And this is *necessary*. For example, studies about the nature and possible forms of intelligence should be ongoing before a discovery, so that there will be work and expertise to draw upon when rapid assessment of evidence of extraterrestrial techno- or bio-signatures is needed. Similarly, the best time to discuss ethical principles, information dissemination and misinformation correction strategies, and best practices/codes of conduct, is

always now, not ‘defer until later, as needed’ [1, 15, 16]

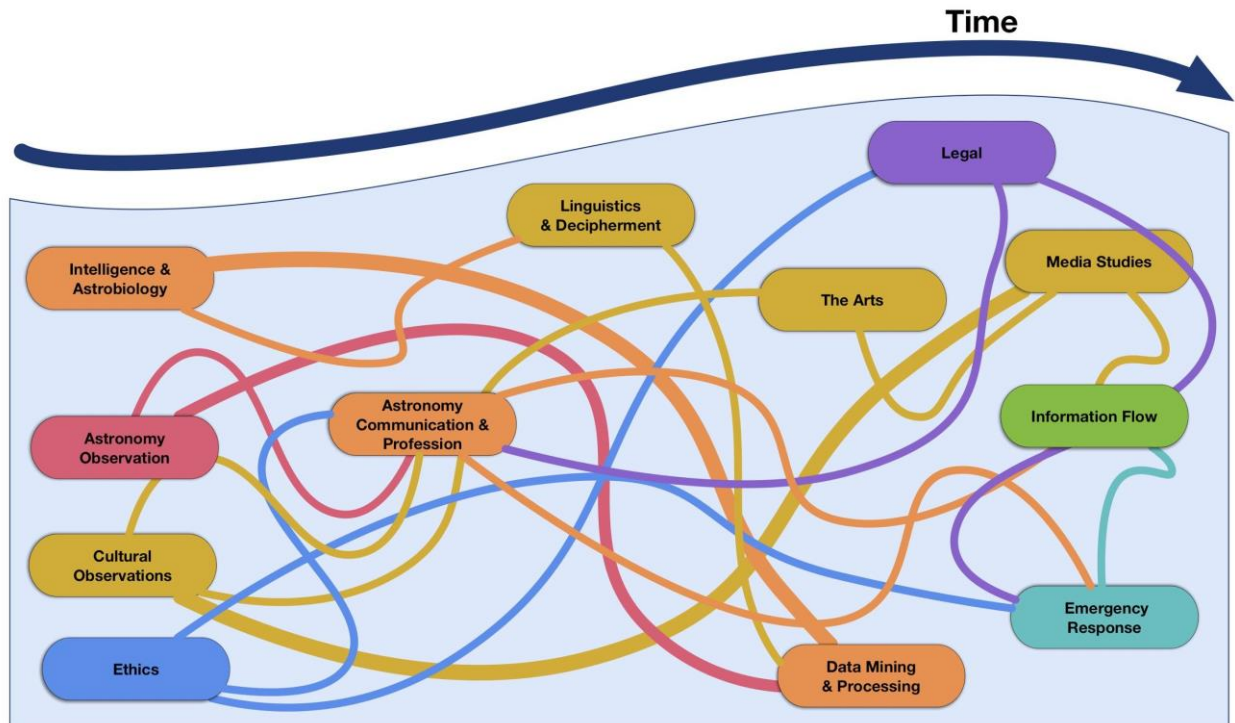


Fig. 1. Streams of work during the process of discovery, connected by weaving, knotted, and overlapping threads.

The connected streams of work are illustrated in Figure 1, a diagram visualizing relationships between selected streams of relevant work as the detection/discovery process occurs over time. Streams of work are depicted as colored blocks within a wave-like shape, with various streams connected by weaving, knotted, and overlapping threads. The tangled nature of the illustration indicates the increasing complexity, non-linearity, relationships, and reconfigurations of relationships that occurs as the detection process unfolds. Time is depicted as an arrow along the top of the wave, progressing from left (present) to right (future) on the horizontal axis. Streams of work in the diagram include: Ethics, Cultural Observations, Intelligence & Astrobiology, Linguistics & Decipherment, Astronomy Observation, Astronomy Communication & Profession, Legal, Information Flow, The Arts, Media Studies, Data Mining & Processing, Emergency Response.

Before and while a discovery unfolds, these streams connect, in various ways, all along the timeline. All expertises are needed, in varying ways and degrees, throughout the process. And when possible detections are announced -- or, eventually, confirmed detections -- then the connections will grow

denser, so the fields effectively pool together as though in rising floodwaters. It is also important to note that culturally speaking, time is not simply linear: the past and future are always with us in the present in “timeknots”, and particularly so at momentous times [17,18]. For example, prior announcements about putative detections affect subsequent reactions by media and publics and well as other scientists. They also constitute a valuable source of data about information flow, scientific debates, and public reaction.

2. From past work to a future research roadmap

2.1 History of interdisciplinary contributions

There have been many conferences and workshops in which historians, psychologists, anthropologists, and other scholars have met with astronomers, planetary scientists, radio astronomers and other SETI researchers to discuss issues relevant to the question of post-detection and SETI and society broadly. This history should be consulted and understood in the course of any new efforts in order to build on the decades of previous work and collaboration represented by the meetings and resulting publications. Previous meetings include: many SETI sessions at

previous IAC meetings (see iaaseti.org/en/meetings for paper and session listings going back to 1999) and some dedicated IAA SETI post-detection workshops; the CASETI meetings in the early 1990s; the interdisciplinary CONTACT conference (celebrated its 30th anniversary in 2018); the 2010 Royal Society meeting on detection and the consequences for science and society; many relevant papers presented at AbSciCon, Bioastronomy, and Origins conferences; SETI Institute events; sessions at the AAAS; and others including the “Communicating Across the Cosmos” meeting (2014), the “Decoding Alien Intelligence” workshop (2018); the Library of Congress workshop which led to Dick, ed, 2015 [19]; “Making Contact” at Berkeley (2018, 2019), Social and Conceptual Issues in Astrobiology meetings (2016, 2018), and meetings at Green Bank in 2010, 2019. This list is certainly not exhaustive. Table 1 provides a partial summary.

Existing interdisciplinary books and collected works on the subject include these -- and there are more: After Contact: The Human Response to Extraterrestrial Life [20]; Social Implications of the

Detection of an Extraterrestrial Civilization [21]; When SETI Succeeds: The Impact of High-Information Contact [22]; the SETI 2020 Roadmap [23]; Cosmos & Culture [24]; Searching for Extraterrestrial Intelligence [25]; Contact with Alien Civilizations: Our Hopes and Fears about Encountering Extraterrestrials [26]; Communication with Extraterrestrial Intelligence [27]; Civilizations Beyond Earth: Extraterrestrial Life and Society [28]; Solving Fermi’s Paradox [29]; Astrobiology and Society in Europe Today [30]; The Detection of Extraterrestrial Life and the Consequences for Science and Society [31]; Astrobiology, History, and Society: Life Beyond Earth and the Impact of Discovery [32]; The Impact of Discovering Life Beyond Earth [33]; Science, Culture, and the Search for Life on Other World [34]; Archaeology, Anthropology, and Interstellar Communication [35]; Astrobiology, Discovery, and Societal Impact [36]; Starstruck: Cosmic Visions in Science, Religion and Folklore [37]; What We Know About Extraterrestrial Intelligence: Foundations of Xenology [38].

Meeting	Date & Location	Sponsors	Results & Publications
CONTACT Conference	Since 1988: annual then biennial	CONTACT	COTI Contact Simulations
AbSciCon (Astrobiology Science Conference)	Biennial since 2000	NASA, Blue Marble Space Institute of Science, ELSI (Earth-Life Science Institute), Meeting Host Universities, SETI Institute, others.	Abstracts in issues of Astrobiology journal.
IAA SETI Permanent Committee	Versions of the committee have existed since early 1970s, meetings held within annual IACongress, various international locations		Multiple special issues of Acta Astronautica; IAF online archives for IAC meetings; and IAA SETI meeting minutes https://iaaseti.org/en/
Cultural Aspects of SETI (CASETI)	1991-1992, Chaminade Conference Center, Santa Cruz, California	NASA	John Billingham et al., eds., <i>Social Implications of the Detection of an Extraterrestrial Civilization</i> (1999)
Many Worlds	November 22-24, 1998, Lyford Key, Nassau, The Bahamas	John Templeton Foundation	Steven Dick, ed., <i>Many Worlds: The New Universe, Extraterrestrial Life, and the Theological Implications</i> (2000)
When SETI Succeeds	1999, Hapuna Prince, Big Island of Hawaii	Foundation for the Future	Allen Tough, ed., <i>When SETI Succeeds: The Impact of High-Information Contact</i> (2000)
Societal Implications of Astrobiology Workshop	November 16-17, 1999, NASA Ames	NASA	Albert Harrison et al, eds <i>Workshop on the Societal Implications of Astrobiology</i> (1999)
Exploring the Origin, Extent and Future of Life	2003 American Association for Advancement of Science Washington, DC	NASA/ American Association for the Advancement of Science	Constance Bertka, ed., <i>Exploring the Origin, Extent and Future of Life</i> (2009)

Astrobiology: Expanding our Views of Society and Self	May 2008 Univ. of Arizona Biosphere 2 Institute	University of Arizona	Chris Impey et al, eds. <i>Encountering Life in the Universe</i> (2013)
Astrobiology and Society	February 2009, SETI Institute	NASA Astrobiology Institute	Margaret Race et al., “Astrobiology and Society,” <i>Astrobiology</i> , 12 (2012), pp. 958- 965
Astrobiology and Society	February 2009, SETI Institute	NASA Astrobiology Institute	Margaret Race et al., “Astrobiology and Society,” <i>Astrobiology</i> , 12 (2012), pp. 958- 965
The Detection of Extra- terrestrial Life and the Consequences for Science and Society Satellite Meeting	25-26 January, 2010, Royal Society in London, Kavli Centre, Buckinghamshire	Royal Society of London	Dominik and Zarneki, eds., <i>Philosophical Transaction of the Royal Society of London A</i> , vol. 369, issue 1936 (2011)
OZMA@50	Green Bank, WV	NRAO/AUI, NSF, NASA West Virginia Space Grant Consortium	Video archived by NRAO www.gb.nrao.edu/OZMA@50/about.shtml
The History and Philosophy of Astrobiology	September 27-28, 2011, Ven, Sweden	Pufendorf Institute for Advanced Studies, Lund University, Sweden	David Duner et al., <i>Astrobiology</i> special issue, vol. 12 (2012); David Duner, ed., <i>The History and Philosophy of Astrobiology</i>
Preparing for Discovery	September, 2014 Library of Congress	NASA/Library of Congress	Steven Dick, ed. <i>The Impact of Discovering Life Beyond Earth</i> (2015)
Communicating Across the Cosmos	November, 2014, SETI Institute, Mountain View, California	SETI Institute	https://communicating.seti.org/
Social and Conceptual Issues in Astrobiology 2016	September, 2016 Clemson University	Clemson University	Kelly Smith, <i>Social and Conceptual Issues in Astrobiology</i> , in press
Social and Conceptual Issues in Astrobiology 2018	April, 2018 University of Nevada, Reno	University of Nevada, Reno; Blue Marble Institute; others	Ted Peters (Ed.) <i>Science and Theology</i> special issue, Vol 17 (2019); Kelly Smith and Keith Abney (Eds.) <i>Futures</i> special issue, in press
Decoding Alien Intelligence workshop	March, 2018, SETI Institute, Mountain View, California	SETI Institute	Participant White Papers (daiworkshop.seti.org)
Making Contact 2018	May 2018	Berkeley SETI Research Center	Working Group Statements, Making Contact Participants (Oman-Reagan & Webb, eds. In preparation.)
Moonshots and Earthshots in the Search for Life Beyond Earth	Green Bank WV, 2019	Unither Bioelectronique, Green Bank	Talks online at https://greenbankobservatory.org/science/ meetings-and-workshops/moonshots-and- earthshots/

Table 1. Some Relevant Meetings, Workshops, Conferences, and Publications. (Adapted from Dick 2018 [7])

2.2 Moving from historical analogies concerning contact to a present-oriented role for social scientists and humanists

See Section 6 for remarks on the use of analogies to explore the scientific discovery process; here we comment on historical analogies for actual contact.

In the earlier days of SETI, there was considerable emphasis on the possibilities for actual contact between human and extraterrestrial cultures, with transmission of information. (Other discovery scenarios were less considered.) This is one reason why historians and anthropologists were interested and involved: prior episodes of contact between different

human cultures on Earth were thought to be potentially informative. Such human-human contact episodes are informative about a wide variety of potential futures involving SETI, but perhaps primarily in a different way: they highlight ethical errors of the past, and ongoing patterns of colonial thinking and behaviour which we would do well to consider. As Denning [39] notes, human history has frequently appeared in debates about SETI public policy, particularly in reference to the history of contact between cultures on Earth. However, these analogies rely on accounts of contact which are themselves often partial, contested, and significantly biased. The popular versions of world

history SETI researchers sometimes turn to for historical examples of contact are often “deeply politicized, significantly slanted, and tend to reduce complex truths to simple stories” [38; also see 40, 41, 42].

In the present paper, we do not address potential long-term results of contact, should it ever occur, and instead focus upon issues concentrated in the present and near future which pertain to scientific practice in society. We are thus bringing social science and humanist expertise to bear on issues of more *proximate* concern, and suggesting that we focus on diverse recent and current datasets about human behaviour. (Note: others have used this approach before, e.g. Harrison 1997 [20].) We focus here on known and knowable human-human, human-technology, and human-information interactions, rather than upon hypothetical human-alien interactions.

2.3 Recent Efforts concerning post-detection

The present authors have each been active in relevant discussions, projects, and publications. A brief review of some recent activity follows.

2.3.1 Towards an IAA-SETI protocols revision:

The IAA SETI Committee advanced the discussion of post-detection issues with the drafting of the Declaration of Principles Concerning Activities Following the Detection of Extraterrestrial Intelligence in 1989. This Declaration was followed in 1995 by the Draft Declaration of Principles Concerning Sending Communications With Extraterrestrial Intelligence. These documents were prepared by individuals associated with most of the institutions conducting work in SETI, and were met with widespread approval, including being presented to UN COPUOS by Jill Tarter and Ernst Fasani. In 2010 the protocols were revised by the SETI Committee; however, the new version was not acted upon in terms of obtaining institutional acceptance, and in any event quickly became outdated as online news and social media attained prominence. In 2017 Duncan Forgan and Leslie Tennen were tasked by the SETI Committee with conducting a review and potential revision of the protocols. Thereafter, a preliminary strategy was developed for the structure, information gathering, synthesis, and professional and public comment opportunities, that included establishment of a website for the collection of data and feedback. Presentations were made both to COSPAR [43] and the IAF [44] in 2018, and the strategy envisioned receiving institutional and other support, but none could be secured, and the project has not yet been realized. Moreover, while the preliminary strategy received

general acceptance in part of the SETI community, there was some criticism of methodology, giving rise to efforts such as the present endeavor.

2.3.2 Rio Scale 2.0

In 2018, Forgan et al. [45] undertook a revision of the original Rio Scale by Tarter and Almar [46], updating it for present circumstances. Rio 2.0 [45] has particular strengths in the classification of detections which are not very noteworthy or credible, i.e. which score a “0”: this is a valuable contribution in our present day context of sporadic detection claims which are rapidly and highly publicized before peer review, confirmation, etc. Aspects of Rio 2.0 which have sparked discussion include the use of a reference range which may not be optimal for public understanding, and the possibility that the Rio Scale’s ‘impact ratings’ may reflect the interests and concerns of astronomers more than those of interested publics [47,48]. There appears to be a clear role for social science contributions in this regard. The present roadmapping effort may ultimately yield additional ways of effectively communicating SETI results for the benefit of a non-specialist audience, and/or contribute to further refinements of the Rio Scale.

2.3.3 NASA Technosignatures Workshop Report

Although post-detection matters were discussed only briefly at the NASA Technosignatures Workshop in Sept 2018, the subsequent collectively authored report emphasized the need for research that would enable the interpretation of technosignatures findings, post-detection concerns, and funding structures for interdisciplinary work which bridges the sciences, social sciences, and humanities [15].

2.3.4 Astro2020 Decadal Survey White Paper, “Preparing for the Discovery of Life Beyond Earth”

In this “State of the Profession” white paper for the American National Academies Decadal Survey on Astronomy and Astrophysics (Astro2020), Denning and Dick [1] argued that the National Academies should “created a federal funding structure -- e.g., a NASA collaboration with NSF(SBE) and/or the National Endowment for the Humanities -- to foster sustained collaboration between scientists, social scientists, and the humanities addressing ongoing, emerging, and future societal issues resulting from intentional searches or serendipitous discoveries of extraterrestrial life.” They particularly note that “societal impact will very much depend on the discovery scenario, ranging from microbes to intelligence and via biosignatures, technosignatures, or fossils”, and that “the determinants of public response and societal impact are cross-cutting and will involve the multiple characteristics of the detection, including

factors such as certainty, potential threat or benefit, scientific source, nature of the announcement, media response, and so on.” They review the history of NASA’s early support for interdisciplinary work on societal impacts, and the current status, including the absence of a university center or strong and stable network to build upon and extend the interdisciplinary work already done. They outline a rationale for government support resting upon: responsibility to the public; new issues arising through scientific progress; and responsibility to scientists, students, and citizen scientists. They highlight the need for a full research roadmap, and then for research which follows through on the areas defined, support for interdisciplinary education, a centralized archive/repository for this work, support for social scientists and humanists participating in relevant science conferences, and carefully structured adjuncts to the new emerging astrobiology funding/research coordination structures.

Notably, this paper received 119 cosigners/endorsers from the global SETI and astrobiology community in the space of 24 hours, with additional cosigners sending their support after the deadline.

2.3.5 Additional recent efforts: surveys and workshops

Further relevant recent efforts in which subsets of the present authors have been involved include a survey at the Royal Society summer exhibition 2019, meetings of the UK SETI Research Network, papers at IAC meetings, presentations at COSPAR in 2018, the Decoding Alien Intelligence workshop in 2018, Making Contact 2018, AbSciCon 2019, and the Earthshots and Moonshots workshop at Green Bank in summer 2019. Cumulatively, these efforts have underscored the need for improved interdisciplinary communication and strategies for synergy in research.

2.4 Related roadmapping

Research roadmapping is a time-honoured tradition in astrobiology and SETI/technosignatures research. Existing, recent models of roadmaps upon which we might draw include the NASA Astrobiology science roadmaps [49,50,51,52]; NASA Technosignatures Workshop report [15]; Decoding Alien Intelligence workshop (daiworkshop.seti.org), Astrobiology & Society roadmap, Astrobiology and Society in Europe Today, Dick ed. 2015, and Vakoch ed. 2013. Models from the social sciences, particularly those incorporating elements of citizen science and crowdsourcing, also may provide inspiration.

2.5 Starting Points Today

Guidelines for the IAA SETI Committee and the larger SETI/technosignatures/astrobiology community

might usefully be reshaped. The present Protocols require reevaluation, and it may be that a tiered approach will be the best way to incorporate all the necessary elements, e.g.

- a. timeless set of Principles concerning intentions/aspirations within the SETI community
- b. ideal Codes of Conduct (transparency, open-source data, skepticism, multiple verifications, responsible communication with the world, etc.), and
- c. evolving sets of Best Practices (e.g. social media safety guidelines for researchers), each of which would be produced by specifically qualified research teams (e.g. social media; public understanding of science; emergency communications; graphics) and updated regularly to keep up with changing situations.

This could allow the flexibility necessary to react to the details of changing circumstances, and incorporate multiple expertises, while maintaining a core set of ideals.

Any realistic roadmapping exercise of post-detection terrain must acknowledge starting points such as these:

- The contexts of technosignatures and biosignatures research are continually evolving
- The potential forms of detection are continually evolving and expanding
- there is no mechanism for ensuring compliance with Protocols and even if there was, it's possible that some wouldn't comply when faced with real-world pressures/motivations
- there is no mechanism by which to prevent people making premature announcements
- there is incomplete agreement among scientists on what would constitute confirmation of a discovery of ETI, and the process could be prolonged (and the world doesn't wait)
- METI (de novo or reply) cannot be regulated effectively [53]
- we cannot prevent misinformation being spread
- we cannot fully protect scientists, citizen scientists, or students, from online or real-life abuse (doxing, swatting, worse) -- and the law offers only limited protection (see Legal Issues, section 9 below)

2.6 Towards a new research roadmap

The present exercise takes all the above into account in a general way. Below, we address the larger

context, and some potential roadmap areas – some in brief and some in detail. This is preliminary work.

3. The Larger Context: Current Concerns

Much has changed in recent years, because of the searches which have been accomplished so far, and because of the expansion of potential detection scenarios [54, 55 56], but also because of evolving global contexts for this research. This section makes brief observations concerning science and social responsibility, the issue of trust, and existential risks.

3.1 Science, technology, social responsibility

The subject of scientists' and engineers' responsibilities to society is vast and venerable. Unsurprisingly, there are disparate notions within the SETI and astronomy communities, as with any scientific community. What makes SETI unusual and, perhaps, held to unusually high standards, is the historically common claim that a detection will be a boon to all humanity. (Not implying that this claim is made by all SETI researchers.) Social scientists and humanists, with their emphasis on human diversity and particularities and library of illustrations that humans tend not to agree on much, generally consider that the consequences could be mixed. Clearly, scientists cannot anticipate everything that others will do with their scientific results or technologies – all the cultural reactions, manifestations, politicizations, good and bad, etc. – nor should they try. Equally clearly, it's unreasonable to be certain that there would be no negative effects for anyone anywhere in the event of a detection. We are working, therefore, in the middle ground of trying to anticipate and minimize *some* problems. This has been the norm for most of SETI's history [1], going back to the interdisciplinary work fostered by John Billingham and others [1, 57].

There is no clear consensus within the international SETI community regarding scientists' and engineers' responsibilities to society as collectives or as individuals. E.g. While some consider that they are responsible solely for doing the best observational astronomy possible, others consider themselves responsible (not wholly, but as part of a community) for an additional array of issues. Equally, other members of society have varying expectations of scientists and engineers, expectations which are sometimes explicit and sometimes implicit.

Social research specifically mapping out these beliefs could thus be of use. An international, interdisciplinary roadmap could be useful in allowing individuals to locate themselves within the spectrum of expectations of scientists and engineers (national, cultural, institutional, etc.), facilitating discussions across viewpoints, and contributing to consensus about best practices.

It may be helpful to situate this discussion within the larger framework of other sciences and engineering endeavours with expected or real broad human implications, to demonstrate how other fields have dealt with these problems, e.g. providing dedicated and ongoing funding for ELSI (Ethical, Legal, Social Implications) or ELSA (Aspects) work in conjunction with the science itself (see genome.gov for example), and/or enabling citizens to engage in participatory governance. n.b. Labels change, e.g. in Europe the terminology is now sometimes "Responsible Research and Innovation" [58]. Obviously, the entire discussion must be understood within the context of governmental vs. non-governmental funding [59]. Equally obviously, the discussion must be understood within the shifting context of SETI/technosignatures research: for some, inclusion within astrobiology distinctly implies that astroethics or bioethics concerns become applicable. Finally, it is notable that some scientists and tech pioneers are engaged in high profile interdisciplinary initiatives concerning responsible self-governance in fast-moving sectors of technology and engineering, e.g. bioengineering, AI (e.g. Future of Life Institute, Centre for the Study of Existential Risk, Future of Humanity Institute).

3.2 The Problems of Trust

Trust is an essential roadmap research area: we need research about best practices for SETI researchers and allies in their task of providing trustworthy information about putative or confirmed detections. "Common sense" cannot be assumed to work. Luckily, there is a good deal of research to draw upon. Similarly, elsewhere in the research roadmap: trust breaking and forming is inherently related to the diffusion of information, that can isolate people into clusters of the same belief. This is not simply a social media phenomenon, social media only accentuates it; since time immemorial we have tended to trust our close networks, reconfirm our beliefs, and distrust weak links and out-of-network entities and information (i.e., theory of ingroups - outgroups).

But the problem of trust is also present in the larger social context for today's socially-minded scientist 60. The following issues are commonly raised in popular discourse:

- Trust in information delivery systems is uneven at best. News media and social media are all subject to considerable scrutiny and skepticism because of their political influence and commercial power.
- Trust in the funders of scientific research is uneven. Governments and philanthropists alike are commonly assumed to have complex agendas.

- Trust in scientific expertise itself is very unstable. For example, as Nichols points out in his sobering essay “How America Lost Faith in Expertise and Why That’s a Giant Problem”, the American populace has recently become notorious for strong and widespread tendencies to actively dismiss expert opinion and science itself, and for the related resurgence of conspiracy theories [61,62].
- Trust in scientists as a group is likewise uneven. Recent Pew polling data in the USA, for example, are encouraging in some ways (e.g., a strong majority of Americans have “a fair amount” of confidence in scientists to act in the public interest) but discouraging in others (e.g., the gap between those who do have that confidence, and those who don’t, is significant and probably stable due to political correlates). This is useful information to have: it isn’t just the average that matters, but the differences, because they suggest different avenues for trust-building. The details matter, too: for example, in that study, most Americans *don’t* believe that scientists are transparent about conflicts of interest [63]. This is just one recent study from one country, concerning a particular subset of scientists (not astronomers); however, similar studies worldwide yield variable and fluctuating results which suggests that a ‘one-approach-fits-all’ strategy for building trust in scientific communications about extraterrestrial life, would not be optimal.

3.3 The larger context of climate change, diversified existential risks on Earth, and our solar system.

When the primary existential risk to the industrialized world was understood to be nuclear war, SETI was perceived as a potential answer to human troubles – a future-oriented, common human endeavor that could, and did, connect astronomers from the US and USSR through mail correspondence and occasional meetings. The possibility of detection of an extraterrestrial society held within it a specific hope of finding evidence that surviving our own ‘technological adolescence’ was possible, and that perhaps we might join a club of peaceful, older, wiser societies and access a torrent of their knowledge [64]. Today, the power of SETI to connect astronomers and other researchers around the world remains -- as evidenced by this very collaboration, and many more like it. The hope of an easy information-rich detection wanes as searches progress with null results [65], but it is still there. However, much has changed in recent years.

Even as those positive elements – of international collaboration and hope – remain, they have become entangled in the complexities of our new millennium.

Our existential risks have proliferated and worsened – climate change and biodiversity collapse are accompanied by potentially hazardous bioengineering, nanotechnology, AI, and weaponized dis/information. Skepticism abounds concerning the dual edges of technologies which outpace regulation. Networks of information flow are ever more complex. ‘Pure science’ – if it ever existed – seems less and less possible in a world where everything is commodified, in ways both obvious and subtle. Cold War governments are now replaced by networks of power in which commercial interests are as prominent as geopolitical ones; we live in a time of hyperwealthy individuals, extremely powerful corporations, and an unprecedented divide between the richest and the poorest on Earth. There is a growing perception that risks are created by a few and borne by the many – who cannot insulate themselves from them, cannot buy their way out of existential trouble as the wealthy can. And those most intensively backing SETI now – in terms of funds, training, expertise, technology development – include exactly the sectors which are seen to be generating some of those new existential threats, and with them, considerable mistrust. On the bright side, these sectors also unquestionably generate high degrees of respect and enthusiasm for the various benefits that they bring to the world – including the ability to mobilize responses to climate change, and work for social equity, etc. But the overall picture is fluctuating, and technosignatures and biosignatures research and researchers can be affected in ways both big and small.

None of this is entirely new. But the overall character of it is. An essential part of this larger context is also a remarkable division concerning humanity’s future in space. SETI began concurrently with the early human spaceflight program, in an era when it was unthinkable that corporations, or nations other than two superpowers, might have direct access to space. Now, the solar system has become a space where multiple conflicting interests will play out – and some soon-to-be spacefarers would like our corner of the universe to be empty of life, a canvas for human experimentation and expansion, whereas others might like it to be lively. A detection of extraterrestrial life, near or far, techno- or bio-, will be one element within this larger context. So will an ongoing silence.

4. Potential Roadmap Areas

A research roadmap would involve many topics and expertises. This list is exploratory and illustrative, not comprehensive. Many areas have been explored

previously; there is current work available for some, but for others, an update is required.

- the problem of ethnocentrism in imagining responses to detection and in contact/post-detection planning
- the risk landscape
- cross-cultural psychology
- affect, e.g. hope and fear, in relation to SETI
- diversity of perspectives among technosignatures and biosignatures researchers
- comparison of the beliefs and concerns of scientists searching for extraterrestrial life, with those of different communities
- positions of religious leaders and communities [66, 67]
- perceptions concerning existing protocols
- the role of the protocols
- protection of scientists in the era of the internet and social media
- the dissemination of information
- mechanisms and institutions for global discussion e.g. IAA, UN, non-governmental organizations such as the IAF or the Secure World Foundation, or a new dedicated organization
- matters of metalaw
- areas of concern for the public – e.g. what questions would people be likely to ask in different detection scenarios?, what types of answers would be most satisfying? n.b. this calls for more open-ended questions than most surveys include, and a broader cross-section of the global public than has usually been involved [68].

Below, we expand on a selection of areas which we regard as being of particular relevance: specific concerns of astronomers; implications of diversified detection scenarios and players; habits of thought in the SETI community; ideas in wide circulation about possible alien characteristics and behavior; ideas in wide circulation about detection and post-detection; insights from public surveys concerning alien life; social media and information diffusion; lessons from prior examples of claimed detections; lessons from mass warnings/risk communication; economic impacts; and legal issues.

4.1 *Specific concerns of astronomers*

There have been many changes in the decades since the last modification of the SETI Post Detection protocols, and it's difficult to overstate the technological and sociological changes that have been made in that time.

The current ubiquity of social media has led to an environment where scientists have become

accustomed to communicating directly with the public. This could be a problem if done in conjunction with announcements associated with biosignature or technosignature research, where individual and societal responses to discoveries are uncertain. Personal safety can be a concern, especially among women and minority researchers. Institutional media accounts may be used to insulate researchers from direct public responses. Social media safety particularly needs to be seen within the context of State of the Profession concerns within astronomy, i.e. concerns regarding equity for women and minority/racialized researchers. See Appendix B below, for commentary and resources on this subject.

The original/existing protocols emphasize the use of the existing IAU Circular system, which connects primarily to other observatories and astronomers. The Circular system is somewhat outdated and poorly maintained. At the time of this writing the links on the IAU Circular web page do not work. The Circular system relies on astronomers monitoring for new Circulars regularly. The Circular system emphasizes connecting to other signatories to the Post-detection Protocols, and the IAA SETI Postdetection group but these contacts don't work consistently, decade to decade. The most recent erroneous announcements of detections of ETI (RATAN-600 described by Maccone 2016 [69] and Klaes [70], and Borra and Trotter 2016[71]) bypassed both the Protocols and the Circular system.

Any system that isn't used gets rusty. A technosignatures-only network would be used so rarely that it would be likely to decay from disuse. This may be part of the problem with the Protocols. It may be a good idea to attach a SETI announcement network to systems that *are* used frequently, for example Gamma Ray Burst, Fast Radio Burst, or Gravitational Wave rapid response networks. It is worth looking into becoming an adjunct to one or more of these system used in astronomy. (Or perhaps also to other systems used in other fields, too, e.g. data science) There are many alert systems now in astronomy to solicit additional rapid observations at different wavelengths: perhaps there is something there that could be used.

There is less recognition of SETI as a professional astronomy activity throughout much of the world. This results in fewer opportunities for astronomers to enter SETI as a field they are able to pursue professionally, without career impacting repercussions. This further adds to the difficulty in building, sustaining, and leading an international professional conversation about detection management.

Anthropologists who work with astronomers know that a variety of concerns about post-detection issues may be articulated in one-on-one interviews and casual

conversations. Ethnographic and survey research would yield information of value.

4.2 Implications of diversified detection scenarios: new players

One of the challenges facing the SETI community is the influx of new players in the field, as new institutions join SETI efforts and as new non-governmental funding sources become common. These institutions may not be aware of the existence of the Postdetection protocols or may not agree to them even if they are aware.

Astronomers collaborating at institutions/programs focused on SETI will likely have their own internal procedures, the details of which may or may not be entirely public. With the increasing variety of players present and future (who knows what will be happening in 10 years), we cannot assume that everyone is operating or will be operating under an explicit protocol. So a gap exists there. A "1-800-Tell-Us" line may not be easily workable in practice given a possible deluge of spurious reports. However, it is possible that a general code of conduct that is well-circulated would have some effect, e.g. "don't make your announcement on Twitter first, even if that's the fastest way to directly connect with your fellow scientists". And perhaps it is worth creating a code of conduct that people would be required to review and signal their acceptance of, as a condition of access to data for citizen science or educational projects. It is also possible that such a condition could be enforced at the observatory level.

We might look to practices in a variety of citizen science projects and participatory governance too - people can and do hold themselves to high standards, but how can this best be encouraged? This doesn't 'just happen' automatically, but requires education of both the project leaders and participants. Some basic education about the diversity and possible range of human reactions to a detection (enthusiasm, fear, anger, worry), might support adherence to these high standards.

Part of the original purpose of the Protocols was to give global (Soviet in particular) colleagues support with disclosing a discovery, during the Cold War. Political landscapes changed over the years, and with each change in players -- e.g. China's new prominence in this field with the powerful FAST telescope -- come new challenges in terms of open communication across cultures and political systems [72]. Anthropological research can be a tremendous asset in understanding worldviews in very different cultures who converge upon a modern scientific project. For example, some cultures may be opposed to open data, e.g. making data available before it has been fully analyzed, potentially resulting in false claims, or in the case of a detection ensuring proper credit to the discoverers.

Astronomers frequently deal with multiple, often implied, codes of conduct when conducting research; these range from acknowledging observatories and software developers, following observatory imposed proprietary data that may only be in the observers possession, to codes promoting inclusion of under-represented groups, and of course the Protocols. One drawback of this wide array of codes is that it is likely that most astronomers don't have a detailed list of the codes of conduct that apply to their work. This can lead to conflict between researchers.

4.3 Habits of thought in the SETI community regarding post-detection and societal impacts

The SETI community is not monolithic. In identifying some habits of thought and expression, we do not assert that everyone involved thinks in these terms, or that they do so all the time -- merely that these are constructs which do appear with some regularity and which may usefully be addressed in efforts to think rigorously about post-detection matters.

- In writing and speaking about "the public", it is easy to forget that this is a fiction of sorts: "the public" is composed of people, in all their diversity of views and beliefs.
- "Scientists" is also not a monolithic category. Scientists are similarly diverse in their views about matters both scientific and human.
- "On average" or "most people" is a type of generalization which may be convenient at times, but if we ignore "the rest" or "outliers", we miss a great deal of relevance. It's the differences between people that generate history.
- Incomplete identification and understanding of cultural, political, sexual, racial, religious and other biases in traditional science and engineering fields.
- Analogy: historical analogies are often used with imprecisions that matter.
- Tendency to reduce the matter of detection/reaction to a "yes/no" answer to the philosophical question "Are we alone?", whereas it's more likely that a detection will initiate a cascade of questions, many of them uncomfortable.
- A conflation of biosignatures and technosignatures (at least in the media and the community's interactions with the media). While they both entail a post-detection scenario, the ramifications of each detection are different and this should be appreciated.
- The myth of the "Eureka" moment. A "detection" will probably require years of verification and follow-up work, and post-detection scenarios will initially rely on very

poor signal-to-noise ratio, and a great deal of skepticism as to whether it is ETI.

- Understandable tendency to think that humanity has always wondered if we are alone, and to reproduce that idea in service of SETI because it is a fascinating question to us - when in fact most of humanity in most of the world has never really thought they were alone, but rather lived in a cultural world with many other kinds of beings, lives, cultures, systems, and even worlds.

4.4 *Ideas in wide circulation about possible alien characteristics and behavior -- the potential impact of these.*

Some SETI researchers minimize their speculation about what ETI would be like, while others speculate freely. There was a time when most SETI literature was unpublished or in limited circulation. That time has passed. Of what scientists involved in SETI themselves say and write/have written, much is very easily accessible online now. The same is true for scientists interested in SETI but not actually engaged in technosignatures or biosignatures searches – and that number is much higher. And a significant proportion of this content is potentially frightening, or at the very least, not reassuring. The ongoing lack of detections continues to provoke speculation about “The Great Silence” and not much of this is cheerful in tone. It is predictable that content such as this would be quoted loudly and often, if a credible detection is announced. Given that there might be little known about the source of a technosignature, there would be plenty of room for speculation [73], and it could be difficult to resolve differences of opinion.

The influence of celebrity scientists – no matter what their field of expertise – could be pronounced. The late Stephen Hawking, for example, often remarked that aliens would be dangerous, and was quoted very widely indeed. In his 2013 article “The Hammer of Hawking” [74], Gazan mapped the influence of Hawking’s 2010 remarks in *Into The Universe* in over 2000 web postings, and found that “The most frequently observed category of agreement was “the hammer of Hawking,” where posters indicated that Hawking’s interpretations were correct, or at least more plausible than others, simply or primarily because they came from Stephen Hawking. These were found almost half the time in conversations among commenters, where one poster tended to invoke Hawking to squash another poster’s dissenting opinion.” Analyses such as these are easier than ever to do, and certainly worth considering.

Further, any detection of extraterrestrial life would intersect with existing public ideas about those forms of life [41,75]; a better understanding of those cultural

landscapes is necessary, to move this discussion forward [76].

4.5 *Ideas in wide circulation about what would occur immediately after detection, and their influence*

Despite there being no singular “public” as mentioned previously, there is a powerful global business of distributing entertainment products including films, video games, and other media. Depictions of extraterrestrial contact in this media, films such as *The Day the Earth Stood Still* (1951), *Contact* (1997), *Arrival* (2016), and others [77], dramatize various forms of response to detection or contact. These depictions provide widely available examples for public consumption modelling how government and other authorities may organize a response, how they might coordinate internationally (if at all), whether scientists or the military would be the authority overseeing the detection, and often even discuss specific ideas about post-detection protocols. These media productions also depict a wide range of imagined public reactions [78], often telling viewers a story in which people panic, create new religions, shoot guns at the sky, hide in bunkers, or become suddenly enlightened or enslaved. While SETI and astrobiology often struggle to distance the scientific search for life from science-fiction and UFO culture, studying those areas from a social scientific perspective offers opportunities for understanding public perceptions and ideas in circulation about detection [79,80].

Examples of speculative detection scenarios in science-fiction are another way in which we see that acknowledging diversity matters for understanding public beliefs and responses. The way detection and contact are depicted varies considerably across and between cultures, and across time in relation to political moments, fashions, and sentiment in popular culture. In the 2017 Russian film *Attraction* (Притяжение), the initial response is solely military jet fighters and missile launches with no communicative effort from humanity. In contrast, consider the 1968 film *2001: A Space Odyssey*, in which the initial detection is discovered by scientists, and the first response is an archaeological excavation and scientific fieldwork. There are also key differences, for example, between depictions of contact in Indigenous science fiction vs. euro-american big budget Hollywood film [81].

In addition to entertainment media, reporting on the results of polling and surveys of public attitudes also constitute a form of influential media in this space, through both news media narratives about the surveys and the use of the results.

4.6 *Insights from surveys: a window into diversity*

Recent survey research concerning public reaction to hypothetical extraterrestrial detection scenarios underscores what social scientists expect: that there is diversity in public opinion concerning extraterrestrials, and that this diversity has patterns which are meaningful. Weigel and Coe [68] review some relevant surveys before 2013, emphasizing that these studies were generally targeting relatively well-educated Western countries, with the one exception of Vakoch and Lee 2000 [82]. More recent surveys of various scales include: a 2017 global survey by Glocalities [83], a survey at the Royal Society summer exhibition (2019, analysis pending); Kwon et al's 2017 study, which went beyond questionnaire format [84]; and a 2015 survey by Yougov. Surveys provide a useful 'map' of beliefs – within the parameters of the survey's sample (which may or may not be representative of a larger population), within the questions asked, and within the question design (e.g. may force yes/no answers and preclude 'don't know'). However, these may not answer the real underlying questions that some SETI researchers have, e.g. 'how should we make decisions after a detection?'

4.6.1 Survey Example: *Survation Sept 2019*

As one very recent example, in a simple survey conducted in Sept 2019 [85] – commissioned by Peter Hatfield (astrophysicist) and Leah Trueblood (legal scholar) of Oxford University and carried out by Survation – a representative sample of UK residents (around 2000 individuals) was polled to gauge public opinions about two aspects of a hypothetical confirmed detection of an unambiguous message from extraterrestrials. First, the poll asked, how should humanity's response to this message be determined: by a team of scientists, by elected representatives, by planet-wide referendum, by a citizen's assembly of randomly selected adults, or 'don't know'? Second, the poll asked if a planet-wide referendum was held, would the respondent vote to initiate contact, not to initiate contact, abstain from voting, or 'don't know'? The answers are quite striking in some of their variations between the sexes, across age groups, education levels, income, and political voting history. A few examples:

- 47% of female respondents voted to initiate contact, vs 65% of male respondents
- 67% of those aged 55-64 voted to initiate contact, vs 51% of those aged 18-34
- those with higher income and more education were significantly more likely to vote to initiate contact
- 35% of female respondents wanted to leave the decision to scientists, vs 44% of male respondents

- 46% of those aged 55-64 wanted to leave the decision to scientists, vs 35% of those aged 18-34

Obviously, these results can only be interpreted within the historical context of this moment; with the Brexit debacle currently at crisis proportions in the UK, it is unsurprising that elected representatives and a referendum were not popular options for decision-making. However, scientists should not be too comfortable about their rather narrow margin of victory: 39% of respondents saying that the decision should be left to scientists while 23% said "don't know", does not constitute a resoundingly strong mandate.

Equally, we might wonder how seriously to take 56% of respondents saying that they personally would vote to make contact with the aliens, when "don't know" (21%) is probably the most reasonable response when the questioner has given them no information about the hypothetical incoming transmission. Or, more to the point... we might wonder what would really happen in such a referendum: i.e., what questions people would have before voting, what information would be provided to them, by whom, and to what end. (See above regarding Brexit.)

Most of these surveys mentioned above was undertaken partly in relation to public outreach exercises. There are inconsistencies in survey design which mean the datasets are not comparable. Further, there are multiple limitations and caveats here which point to areas for future research.

- It's common for these survey results to be distilled in headlines or quotes to a point that elides all the complexity and may even be misleading: what effect does this have? Interested technosignatures researchers need to be aware that it isn't the "on average" answer that matters, and that the details of survey methodology and sampling etc. are *always* relevant.
- But further, these surveys don't go deep enough: there is also the considerable problem of more realistic scenarios. For example, how would different sectors of the public react to a scenario where there is little known about who/what created the technosignature, i.e. if researchers couldn't answer their ostensibly simple questions?
- It is by now obvious (and unsurprising) that consensus on matters involving extraterrestrials is hard to achieve, both in terms of public opinion and in terms of expert opinion. Surveys will not simply tell us what to do.

The majority is not always correct or prudent. And an emphasis on simple majority opinion, it should now

be evident, is a recipe for discontent among the marginalized. The best approach may be to simply learn from the diversity of responses: appreciate what it says about people's ambivalences and confidences, and understand where we ourselves fall on a spectrum of opinion. This can be an excellent starting point for conversations that matter.

From an anthropological perspective, these numbers are just a beginning, a way into some important questions, such as: why *exactly* are there differences between male and female respondents, between age groups, across political affiliations, income, and education? Are there additional differences within those categories (e.g. language, religion, ethnicity, racial identity, cultural background) which also drive the patterns? What does this tell us about science and these members of the public?

There is yet another layer to consider with surveys. For example, given that humans are notoriously poor at affective forecasting (i.e. knowing how they will feel about something in the future) [86,87], then what, if anything, do surveys such as these tell us about how people would *really* feel and react if the hypothetical circumstances became actual?

Additional methodological issues abound: those who work in the human sciences know that surveys have many limitations, but specialists in other areas may not be fully aware of all the issues, so any research roadmap should carefully delineate these. For example: question phrasing can heavily determine responses; affective forecasting limits the literal predictive value of surveys; the 'mischievous responder' effect can be a substantial problem, e.g. with teens online [88]; paid survey-takers; the 'hypothetical bias', i.e. difference between stated values versus real values, has generated a huge body of research [89]; the Hawthorne effect i.e. the effects of being studied [90]; normative expectations, i.e. once one has been told what one should believe, one is more likely to report believing that... even if one doesn't. All this is precisely why 'big data' is increasingly employed to study behaviour (and manipulate it) [91]. And it is also a reason for 'mixed methods' approaches in qualitative research, e.g. also using focus groups, interviews, simulations, as well as linguistic analysis [84, etc.]

If our real interest in surveys is that scientists wish to involve other people, meaningfully, in decisions that involve everyone, we might look at other bodies of research. For example, we could look to new scholarship in the science of deliberate democracy, which "seeks evidence on the capacities of citizens as they engage democratic dialogue, not as they respond as isolated individuals to survey questions" [92].

5. Social media

Social media is part of the detection/discovery picture in multiple ways, as flagged by C. Oliver [2010, 93], Jones [94], and others since. Contemporary detection and post detection scenarios will unfold in the complex (global) sociocultural context. The science teams tend to be interconnected networks rather than isolated groups of expertise. The Internet makes it possible for the information about the detection to be shared in real time via digital platforms and social media. More emphasis is hence put on the scientists and/or engineers as key actors in the scenario and their ability to handle the situation. The new ways of social interaction will shape the information diffusion and also enhance the cultural adaptation. There are risks inherent in this picture, for example, the problem of "digital wildfires" -- "viral spread of information that is either intentionally or unintentionally misleading or provocative" -- which constitute a new form of risk. The WEF accordingly encourages an "ethos of responsibility" for generators and consumers of social media [WEF 2013].

Elsewhere in this paper, we have noted the necessity of safe social media practices, for the protection of those (researchers, students, citizen scientists) involved in technosignatures searches, detections, and announcements. Here we note some additional areas which require considerable research, to map out likely pitfalls and to determine best information dissemination practices in discovery scenarios. This area of study is vast and growing rapidly; here, we build on previous work [95,96] and only scratch the surface.

As always, the larger context here is salient. Social media platforms and corporations have recently been under intense and sustained scrutiny internationally, because of: privacy breaches with users' data; the characteristics of 'surveillance capitalism' associated with these platforms, i.e. the monitoring and use of behavioural data for profit [97]; political influencing including election manipulation [98]; and issues of content moderation by the platforms themselves [99,100].

Of particular relevance to our roadmapping: Crucially, social media is not simply a way of communicating information, unaltered, from a scientist or journalist to the end user. Much social media content is user-generated or user-edited. Obviously, this reduces the original content provider's ability to maintain accuracy. But to complicate matters even more, inaccuracies may spread faster and further than the truth. In "The spread of true and false news online", Voutsoughi et al. reported on their massive study of Twitter rumour cascades in 2016-17. They observe: "False news reached more people than the truth; the top 1% of false news cascades diffused to between 1000 and 100,000 people, whereas the truth

rarely diffused to more than 1000 people. Falsehood also diffused faster than the truth” [101]. Related analyses suggest that on Facebook, the spread of misinformation is patterned according to partisanship, ideology, and age, and that a very small number of users were responsible for a very high proportion of ‘fake news’ dissemination [102]. The details will vary according to the subject at hand, but studies such as these indicate the need for sustained attention to phenomena of misinformation [103], and phenomena of information flow.

Existing models of “information diffusion”, from mathematical to computational models, can be used to examine the patterns of *how* information spreads, *which* information spreads (for example, virality as some models mirror epidemiological models), *where* nodes or bottlenecks are, network topologies, and how this information or misinformation has effects in the real world [104]. These models are relevant not only for understanding human biases and the spread of misinformation, but also for disaster and crisis management.

Models of information diffusion may also offer tools to understand which information online is noise without real impact. Memes and social media trends may be transitory and have a short half life, but information that stays may have a potential to cause damage -- to accentuate confirmation biases more than inform or clarify. When considered as language, information appears to spread in clusters in power law types of distributions, where one quantity varies as a power of another. The important information may be buried in the noise, while also being most impactful. The phenomenon of information movement can have a fluid-like dynamic (known as statistical mechanics in computer science). Much like weather, information movement can appear chaotic, leaving us struggling with the challenge of understanding its effects and intervening to manage its impact [105].

Detection scenarios could be explored using current computational methods and theories of information diffusion in society. While there are significant events that can be influenced both positively (i.e., rescue efforts) and negatively (i.e., misinformation for business gain), by the spread of information online, a detection scenario would be complex enough where the information spread online may be only a small part of what influences decisions and actions at the individual level. Ties between family, friends and neighbors could be more influential on the level of impact. The possibility of less confidence in the future could potentially lead to economic crises but also to new business opportunities.

Several studies reports a report a strong tendency to disseminate negative content to more people, to focus

on that negative content for longer, and to convey negative information in more elaborated and detailed ways. In a review of literature Hornik et al. [106] found the “negativity bias” in word of mouth communication was often explained by negative information being perceived as more thought provoking, more surprising, and more influential and trusted. In a broader 17-country, 6-continent study of “psychophysiological reactions to real video news content,” Soroka, Fournier, and Nir [107] also found negativity bias in reaction to the news. They concluded that such bias is not a uniquely US phenomenon, and there is a “a mean tendency for humans to be more aroused by and attentive to negative news.” They also found a great deal of individual variation on that mean.

As of July 2019 3.5 billion people were actively using social media, 46% of the global population. Over the last few years India, Indonesia, and Brazil often join the United States in the list of top four countries by highest number of Facebook users (based on data from We Are Social, DataReportal, Hootsuite, and Facebook). While social media use has plateaued in the Global North (often called “developed” countries), it continues to rise in the Global South (often called “developing” countries) [108].

At the same time online space can spread ideas transnationally, it can modify those ideas and create new meanings and uses [109]. A particular word or phrase might mean primarily one thing in the place of origin, but adopt entirely new meanings as it migrates across the web and across the world. Online space can have the effect of speeding up the creation and spread of multiple meanings, which include crowd-sourced contestations of scientific authority with surprising cultural currency. The rise of the “flat earth” movement online is one example of this, an idea that has gained popularity in part due to proponents’ strategic uses of Facebook and YouTube. The flat earth movement provides insights into what these kinds of movements may actually be about, namely defining authority, casting off stigma for “wrong belief” not the “spurious expression of an antiquated worldview” as is often assumed [110]. Actions, motives, and meanings are as complex in online spaces as they are in other spaces and shouldn’t be overly simplified based on assumptions. Instead it’s possible to study how ideas move and change online, and there is a large body of research on this issue available now that was not available just five years ago.

Ethnographic studies on social media use conducted by the UCL Global Social Media Impact Study (ucl.ac.uk/why-we-post), an anthropological research project looking at social media use in Trinidad, Brazil, Chile, England, Italy, Turkey, India, and China, may offer insights into how social media use varies and whether concepts (like negativity bias

in word of mouth information transmission) can be applied to different cultural contexts of social media use, or not.

Extraterrestrial life is an extraordinarily compelling subject for many people, which has additional implications. There is abundant misinformation already in circulation, some of it for decades. We don't have to just *expect* that anything involving aliens would spawn conspiracy theories. This principle has been *demonstrated* amply already. The topic lends itself extremely well to clickbait, particularly because of the information gaps which drive curiosity [111]. It may also attract the attention of people in high governmental office who use social media for political purposes and tend to represent their own opinions as fact [112]. The dissemination of "deepfakes" is increasingly a concern and it is entirely plausible that these would circulate after an announced detection, even if only in jest. (The current tendency for technosignatures and astrobiology meetings to be available in livestream certainly provides a plethora of raw material.) Further, online jokes concerning extraterrestrials can demonstrably spin into real-world phenomena with strange impacts: e.g. an ironic suggestion on Facebook that people should storm Area 51 en masse, "[Storm Area 51, they can't stop all of us](#)" resulted in a viral movement and over 2 million "likes", warnings from the FBI and the Air Force, and, eventually, several real-world events in nearby locales with a variety of economic and environmental impacts, including a need for heightened security and emergency personnel [113].

Further areas for extended research include:

- imagery in news sources concerning extraterrestrial life, particularly given that the amount of time/attention that readers spend reading science journalism can be incredibly short (N. Drake, comment at Green Bank workshop July 2019)
- studies of word-of-mouth or conventional media (e.g. radio) information transmission which may have relevance for populations with variable literacy and information access, and/or with disrupted access in times of crisis. The example of lethal misinformation in the Ebola epidemic in DRC is salient. Notably, social scientists have been instrumental in mitigating the worst of it [114,115,116].
- studies of information spread in different demographics, including children and adolescents. Most studies are geared towards adults, but they are not the only people who matter.

Taken together, research in these areas would help to suggest best practices for maximally clear information dissemination in the age of Twitter,

Facebook, Instagram, and more, and would contribute to defining best practices for reducing targeting of scientists on social media.

As Rick Weiss, Director of SciLine at the American Association for the Advancement of Science, Washington, DC notes, it is an increasingly well-recognized reality that there are significant "societal costs of scientific misunderstanding," including "resources wasted trying to correct misimpressions about, say, vaccines or climate change." He adds, "with the risks posed by misinformation rising every day, I encourage everyone in the scientific community to consider how they can cultivate the opportunity space that is.... 'left of wrong,'" i.e. the time period *before* widespread misinformation causes trouble [16].

6. What can be learned from previous examples of claimed detections?

This is a fertile area for research; although a considerable amount has already been written about historical examples, particularly by S. Dick, [7, 117, 118], they remain imperfectly understood by many, and more recent examples may yield abundant insights and guidance for those involved in technosignatures searches today and in the future.

The case of Percival Lowell's 'canals on Mars' is often mentioned as evidence that a detection of extraterrestrial life had little impact upon the public in the past, and therefore there wouldn't be hysteria in the streets today. But this may be too narrow a reading: Lowell's claims (and he was hardly alone in making them [119]) were made at a time when space travel was a less common idea, i.e. what could be seen through the telescope was assumed to be permanently distant... but he directly inspired HG Wells to write *War of the Worlds* a few years later, a work which forever changed that assumption in the popular imagination, and which spawned a genre of sci-fi which has in turn immensely shaped the context of later research and speculation about extraterrestrials. Lowell's inaccurate observations have thus reverberated for over a century in quite real ways. Lowell also, arguably, sparked a considerable amount of Mars research which has directly contributed to modern astrobiology [117].

More recent examples from both astrobiological and astronomical domains include ALH84001 [120], YZ-Cet, EQ-Peg 121, GFAJ-1, Oumuamua, Tabby's Star, HD 164595/Ratan-600, Borra and Trottier [71] and FRBs. Cases such as these (and more...) may be instructive as 'real-world experiments' amenable to multiple forms of analysis [7, 94, Forgan and Scholz 96].

7. Example: Lessons from mass warnings / risk communication

Although a SETI/technosignatures detection would not resemble a tornado or flood in reality, some of the mass emergency warnings/risk communication research has applicable lessons. [See 68]. This is because risk perception rather than actual risk, drives behavior, and related information-seeking-and-sharing behavior is of particular interest to us here. Emergency warning researchers are investigating subjects of great relevance, including effective brief communication about complex and concerning topics, using social media.

There is a substantial international literature on public responses to mobile phone mass emergency warning systems. That literature indicates that considerable research still needs to be done, and urgently – studies of these systems’ effectiveness and impact are lagging years after the systems’ deployment [122]. But the literature also suggests that a person’s perception of the announced threat, their expectations of their ability to cope, social norms, the quality of the message, and the sufficiency of the information provided, are all key factors in determining reaction and behavior [122]. Models indicate that “the receiver of a warning message first assesses the threat level,” and then assesses its personal relevance, and then assesses their personal ability to cope adequately [122]. The latter in turn is based upon their prior beliefs about self-efficacy *and* on the quality of information provided/adequacy of the advice given [122]. Crucially, when a “when the threat is seen as relevant but coping is seen as impossible, some psychological reframing of the situation is a likely reaction” and can include a variety of maladaptive reactions [122]. But also, crucially, the person will be driven to seek more information – and spread it. And the perceived quality/reliability of that information is crucial: “in emergency situations one can assume that people need information that they trust and can adhere to, without doubts” [122]. Information-seeking behavior varies according to factors like age, risk perception, educational levels, social norms [123].

The 2018 National Academies consensus report, *Emergency Alert and Warning Systems: Current Knowledge and Future Research Directions* [124] describes the current complex information ecosystem... “information ecosystem is much broader, including additional official channels, such as government social media accounts, opt-in short message service (SMS)-based alerting systems, and reverse 911 systems; less official channels, such as mainstream media outlets and weather applications on connected devices; and unofficial channels, such as first person reports via social media. Traditional media have also taken advantage of these new tools, including their own mobile applications to extend their reach beyond broadcast radio, television, and cable.

Furthermore, private companies have begun to take advantage of the large amounts of data about users they possess to detect events and provide alerts and warnings and other hazard-related information to their users.”

The report identifies some necessary emergency alert research which may well be transferable to our context of announcing a discovery and using social media effectively:

- “research is needed to understand how to most effectively craft 360-character messages as well as under what circumstances and what message content should be included in linked media. Also unknown is how to craft [alerts] so that they galvanize people to read the entire message, including potentially life-saving linked content.” [124, p 30]
- With “mock WEA messages (90 characters) and mock Twitter-length messages (140 characters), the most common initial reaction was confusion and fear accompanied by a desire to acquire more information about the apparent threat. The lack of clarity about the hazard, the protective action guidance and how long they had to complete it, the time of the incident, and the affected area led to confusion and frustration. Because of the limited information and lack of personalization in these messages, most participants in the study could not tell whether the hazard would impact them specifically. Consistent with public warning response models, participants would have sought out additional information before taking any protective action.” [124, p 35]

The report further underscores that many “Message Receiver Characteristics” affect message reception, including “age, gender, ethnicity, socioeconomic status, race, and disability status.” [124, p 37], and describes the immense challenge of linguistic and cultural diversity within the US generally and large centers (e.g. New York City, with 208 spoken languages) specifically, which “leaves a large portion of the population susceptible to not receiving or to misunderstanding WEA messages”. And many within US borders would prefer to receive alert messages in languages other than English. Further, for some communities, posters and pictures were essential for disseminating information. There are also respondents who rely upon information from family and friends, radio, television, in preference to social media. Also notably, social media users tended to be highly receptive to disaster messages and tend to share them.

7.1 Some lessons for SETI from mass warnings / emergency alert systems?

First, in the event of an announced detection, it is entirely possible that a mass warning system would be used – legitimately, or by rogue actors, or profiteers. Reportedly, some such systems are easy to hack [125]. Second, it's also possible that similar phenomena would arise spontaneously as a result of viral communication / much-followed Twitter feeds etc.

Reactions would vary, but would be heavily determined by the type and quality of the information provided, and the perceived reliability of that information. They could be positive, but they could also be ambivalent or negative. For example, a situation in which scientists can only say “We have confirmed that there is an alien presence out there, but we do not know anything about alien motivations or capabilities” is potentially volatile, particularly given (a) the abundant preexisting well-circulated scientific literature by credible authors speculating about potential extraterrestrial hostility, predatory tendencies, unimaginable power, complete superiority over humans, etc, e.g. [126] and (b) the likelihood that indications of distance between Earth and the location of the signal will be misunderstood by many non-astronomers. In other words, scientists may not be in a position to effectively reassure the public that there is no threat, that any threat is distant rather than proximate, or that anyone has a reliable plan for mitigating threats. The existence of extraterrestrial life would be a fascinating scientific fact to some, but for others, it could be perceived as an amorphous but serious threat. Accordingly, it is important to have the best possible communications strategies.

8. Economic impacts

The World Economic Forum designated the discovery of life beyond Earth as one of five “X-factors” -- “emerging concerns of possible future importance and with unknown consequences” -- in their Global Risks 2013 report [127]. And indeed, in the case of detection of life, whether biosignatures or technosignatures, we can expect some of the effects upon society to also have some significant *economic* costs. Some examples come from the current disaster response literature, and scientific impact upon society literature, as well as from the newer literature on the tangible effects of information diffusion through online networks and traditional media [128]. As we are thinking about detection as a process and not a one-time sudden event, here are some short-term and long-term economic expectations to be considered. These underscore the need for forethought, and responsible communication strategies which minimize risks of social disruption and/or conflict.

8.1 Short-term Economic Impacts

Mass-media, social media and other information/journalistic companies would likely increase coverage, immediately leading to advertising companies boosting their presence. As more government attention would be focused on this issue - for example, the question of a response -- then other issues will be pushed back and delayed. This can result in loss of business and opportunities for the industries that will not get “prime time” focus [129]. Therefore being prepared *ex ante* with rigor and protocols can certainly offset the delays in the day-to-day business at the macroeconomic and governmental levels. Also short-term, there could be public demonstrations, disturbance, or even riots leading to loss of property, and possibly even of life, as well as losses in work days and labor. Recall that 2019 saw *many* major public demonstrations involving hundreds of thousands of people at a time, on matters of national or global concern, requiring additional policing, emergency services, and causing productivity losses. Even completely peaceful large-scaled demonstrations or celebrations incur major costs to governments and local businesses, as well as providing opportunities for profit to some entrepreneurs. (In theory, such demonstrations might even affect astronomy research facilities.)

Also short-term, there could be losses associated with the spread of misinformation, profiteering businesses, and, in general, overall disruptions of the daily business and work lives that can indirectly impact a large population. In addition to the costs of aftermath disruptions, there may be costs associated with unintended consequences – such as rises in insurance risks and premiums, loss of investment opportunities at the local level and market crashes. This can also result in short term shortages of products, as the chains of supply and the markets could be disrupted. The immediate effect of product shortages may lead people to stock up even more on essentials, thus aggravating potential shortages and possibly leading to behaviors such as looting [130].

8.2 Long-term Economic Impacts

Macroeconomic crises are possible. Due to accrued loss of business and work from isolated or not-so-isolated incidents, the confidence of the population and of the businesses in the future might suffer, thus leading to lower interest rates, smaller investments, a more cautious approach to business and investment overall. Also long term, some businesses related to detection-relevant goods and services (e.g., insurance) can potentially accumulate lots of profit and thus become market monopolies and oligopolies with the power to influence governments or drive prices up, at the expense of the consumer [131].

These are only a few potential economic costs that can be envisioned. There are certainly numerous others. An easy study could currently be done to roughly assess these risks in terms of economic impact; ideally, this should be conducted by governmental or inter-governmental research, not by private institutions.

9. Legal Issues

Potential legal issues abound. They include: protection of individuals from online abuse, including doxxing, cyberstalking and defamation; dissemination of information; issues of metalaw.

9.1 Protecting Individuals

Unfortunately, the law offers only limited protections, especially for activities that the authorities deem civil rather than criminal in nature. Individuals involved with the announcement of a confirmed detection will become instant celebrities, and can be expected to be the subject of widespread traditional and social media attention, some of which may be harmful or defamatory. Defamation laws vary by country, and in the US should these instant celebrities seek redress in the courts likely will be required to prove actual malice on the part of a defendant, which could be extremely difficult if not impossible, especially considering the wide latitude in speech guaranteed by the First Amendment. Consideration of legal remedies presumes that the offending party can be identified, located, and subject to legal process, all of which may be elusive, particularly with social media offenders. Individuals should be aware that public attention will invite scrutiny into all aspects of a person's life and background, and that could influence a decision whether to be a part of a public announcement or to remain in the shadows. It can be asserted that employers and educators should have a duty to keep their employees and students reasonably safe online, or at the very least take affirmative steps to not deliberately put them directly in the line of online fire. The absence of current institutional policies and procedures underscores the need for caution and proactive prevention by individuals, rather than assuming that an effective remedy will be available for any problem that may arise.

9.2 Dissemination of information

A primary legal question is whether data is proprietary? Who is entitled to receive the information, and how can unauthorized disclosures be prevented, both before and after confirmation of a detection? The potential for distribution to be limited can be mitigated by widespread scientific consultation during the confirmation process. Would a signal from space be subject to copyright by the discoverer? The discoverer

technically would not be the creator of the work. Would a copyright be dependent on content, or could an undeciphered message be claimed as proprietary with an inherent artistic and other value, analogous to calligraphy? Whether or not the data is proprietary, could its dissemination give rise to liability for false information, or negligently causing a panic, or contributing to others perpetrating false information? These questions are not adequately addressed in existing law, and assume that there will be widespread distribution and access to the information. But the information could be embargoed by governments, in which case the scientists would be faced with the ethical dilemma of whether there is an obligation to defy an embargo if there is indisputable evidence of ETI? Does humankind have a right to know?

9.3 Metalaw

A confirmed detection raises issues of metalaw [132], which are accentuated in any response from Earth. Will we assume that the content of any message we receive is truthful? There must be a certain level of credibility in any communication, otherwise it could be difficult if not impossible to later convince the other party of honest intentions. Nevertheless, whether or not to respond, and the content of any response, including the level of honesty therein, will be among the paramount considerations in the global discussions that will follow a confirmed detection.

10. Back to the future...

Studying the recent history of science, particularly around contentious issues, can provide guidance on what kinds of challenges SETI scientists and their institutions may face as detection unfolds. Looking at recent cases carefully from a social science perspective will enable the SETI community to develop evidence-based preparation plans, and protocols for detection scenarios. This is an area where so-called common sense may not provide the best guidance. It is safest to assume that people *outside* the SETI community will not react the same way as those *inside* the SETI community. It is possible to study this question, and collect both qualitative and quantitative data using a range of methods from several fields and disciplines.

The role of science in society may seem like a foregone conclusion to those of us within science. However, there are many unresolved questions about ethics, diversity, inequality, and other related areas that impact which science is funded, and how that scientific work is then received by various publics. It is possible to study these questions, and there are disciplines composed of experts who are trained on how to do such studies, and in the theoretical underpinnings of how to analyze the results.

11. Summary: Research Priorities and Conclusions

In this paper, we have, first, proposed a new reimagining of ‘post-detection’ into three entangled themes: preparation for detection/discovery, process of detection/discovery, and protocols during detection. Second, we have reiterated the role of multiple expertises in this project. Third, we have asserted that areas not addressed explicitly in this paper, such as ethics [e.g. 133, 134], and studies of the evolution of intelligence, and linguistics/decipherment, are essential parts of a comprehensive research roadmap.

By asking what basic research needs to be done, now, to fill in our knowledge, and guide policy and practice about crucial matters pertaining to discovery and communication, we have explored some clearly relevant domains of research such as social media, prevalent ideas concerning aliens and contact, survey research, mass warning systems, economics, and legal issues.

At the time of writing, our provisional suggestions for priorities are these:

- concerns of scientists directly involved in technosignatures and biosignatures research – we need to learn more about what these are, and continually update that knowledge in conjunction with research about social media and societal contexts, such as:
- information seeking and sharing behavior, including text and images, as practiced by scientists, the general public, governments.
- areas of concern for the public – e.g. what questions would people be likely to ask in different detection scenarios?, and what types of answers would be most satisfying?

Taken together, these will help in the generation of:

- best practices for individual scientists conducting searches
- best practices for supervisors who are training students and citizen scientists
- best practices for organizations running searches
- best practices for funders, in terms of allocation of resources for ELSI/RRI work
- best practices for science communicators
- best practices for social media use – safety for researchers involved in detections and announcements, clarity/comprehensibility for the public, maximum veracity/verifiability

We emphasize that this paper is one *preliminary* step towards a research roadmap. Producing a full research roadmap will involve many more experts, time, and funding. A fully realized research roadmap could in turn could focus and guide social research related to technosignatures and biosignatures in years and decades to come. However, some topics are both obvious and urgent -- those related to safety -- and

work should continue or begin, rather than waiting for the completion of a full research roadmapping exercise.

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Appendix A: How social research is done

We add this Appendix because social science methods and disciplinary practices are often opaque to practitioners in other fields, including physical scientists and engineers. The social sciences include: political science, psychology, sociology, anthropology, economics, and related interdisciplinary fields of inquiry, research, and study. Each involves different expertises and different emphases. While these fields do share some methods, methods also vary a great deal both between and within these disciplines.

Researchers in social sciences often receive training in both qualitative and quantitative methods. Qualitative methods include interviews, working and living alongside research participants (participant observation), focus groups, and other interpersonal data collecting activities involving individuals or groups. Interviews may be formal sit-down style, or informal and conducted in the course of everyday life, they may take place in an institutional setting or in a home or workplace. Participant observation may last from a few months to years or decades, and is conducted in the field. The field site for participant observation may be a community, a workplace, online, or may include multiple sites. Quantitative methods often overlap with qualitative research, with data gathered through both methods informing the analysis. Quantitative data may be collected through survey instruments, big data mining, or during fieldwork or in a lab. Social scientists often also rely on archival methods, turning to historical archives or other repositories of texts, audio recordings, films, or objects. Mixing several methods in one study is a common technique for conducting research in the social sciences. See [68] for additional information for research methods in diverse global contexts.

Social scientists may use many methods to study the diversity, similarity, or variation among different regions, communities, cultures, or behaviours. They may do comparative analysis or look for other kinds of patterns, stories, or cases. Social science research involving human participants is subject to ethics review. Institutional ethics review boards examine research proposals from social scientists to determine the potential benefits and harms to the participants from being involved in the research, and to ensure best practices are followed in research design, data collection, and other related factors. Ethics review policies, procedures, traditions, and practices vary by institution, region, and nation but share a common interest in protecting those who consent to participate in social science research. A key concept in social science is receiving informed consent from those who participate to ensure that the research risks and rewards have been explained, that participants have a chance to ask questions, and that they understand their participation is voluntary.

Social science funding varies by institution, and by region or nation. Competitive research funding is available for graduate student research (for MA or PhD degree programs), through stand-alone grants from research centers, from local, regional, or national agencies, and through interdisciplinary lines on other science grants. In addition to academic funding and professional work in universities and colleges, social scientists also work for government agencies, corporations, and consulting services. Social science firms are increasingly available to conduct qualitative and quantitative research on any question posed by their customers. Businesses are increasingly recognizing the value of qualitative research in addition to the quantitative data they have historically used. Companies employing social scientists, either in-house or through consulting firms, to do qualitative research include: Intel, IBM, Microsoft, Lego, Pfizer, Samsung, Procter & Gamble, and others.

A recent Astro2020 State of the Profession white paper noted that “federal funding generally available to social scientists and humanists involves competition against research proposed about other worthy and urgent causes (e.g. child poverty, social inequalities, environmental adaptation, refugees)”, and thus recommended that the “this work should therefore be funded in conjunction with the science which is bringing the issues to the fore -- just as it was in the earliest days of the NASA-sponsored search for extraterrestrial life” [1].

Appendix B: Social Media Safety for Researchers

In #Foundthem – 21st Century Pre-Search and Post-Detection SETI Protocols for Social and Digital Media”, 2016, Forgan and Scholz [96] identified a series of concerns related to social media use, and make recommendations for social media safety and communications for SETI scientists. Best practices such as these should be further distilled and widely circulated among the SETI community. Despite the obvious importance of protecting personal security, this issue has barely been recognized let alone examined in detail. This is an issue which has implications for all people involved in the search for life beyond Earth and announcements about it, regardless of standing as a professional, student or citizen participant, etc., particularly given that social media use is increasingly part of astronomy (e.g. comments on Twitter use at conferences: www.astrobetter.com/blog/category/social-media/)

A recent survey report concerning online harassment in the USA may be found here: www.pewinternet.org/2017/07/11/online-harassment-2017/ and this helpfully delineates some key issues. However, the issues may be compounded when gender, racial, and social identities are added: an equitable presence for underrepresented voices in science must be balanced with safety. It is essential to find paths forward which allow for the safest and fullest participation possible by all members of the scientific community, and institutions have a very important role to play in this process. This process begins with becoming fully informed.

There is today a larger discussion within science about inclusion and diversity, and ending underrepresentation of women by increasing their visibility – for example, ensuring that women are fairly represented in higher-profile speaking slots at conferences (www.nih.gov/about-nih/who-we-are/nih-director/statements/time-end-mamel-tradition). This is an essential conversation. However, we must simultaneously bear in mind:

- the issues of pervasive and extensive gendered harassment within science and engineering, which is increasingly thoroughly documented in the American context in a recent National Academies report www.nap.edu/catalog/24994/sexual-harassment-of-women-climate-culture-and-consequences-in-academic, and also here www.nature.com/articles/d41586-019-01303-6 and here <https://journals.sagepub.com/doi/abs/10.1177/1461444818781324?journalCode=nmsa>
- the “double jeopardy” problem for women of color in astronomy and planetary science, who face heightened risks of gendered and racial harassment: <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1002/2017JE005256>
- the extraordinary levels of public online abuse targeting females and minorities, which has now been recognized as a human rights issue at the level of the United Nations (2018 report <https://undocs.org/A/HRC/38/47>). See also Amnesty International’s 2018 report, Toxic Twitter www.amnesty.org/en/latest/research/2018/03/online-violence-against-women-chapter-2/ , <https://time.com/5482390/twitter-online-abuse-women-amnesty-international-study/> which documents the seriousness of this issue for women, racialized minorities, and LGBTQA+ people.
- differences in perception concerning the effects and seriousness of harassment online, i.e. women and men, and those in different age brackets, may think differently about this issue, as demonstrated by a recent, extensive American study (www.pewinternet.org/2017/07/11/online-harassment-2017/)

With respect to communications concerning the search for or discovery of extraterrestrial life, we therefore have a difficult tension: for example, highly promoting female and diverse scientists’ voices on social media may serve purposes of equity within science, and be a service to some female readers, but simultaneously carry direct and significant risk to these scientists. One example is the astounding behavior towards a female computer scientist in relation to the 2019 announcement of the first photograph of a black hole, www.theverge.com/2019/4/13/18308652/katie-bouman-black-hole-science-internet. This is not “just one case”. The experiences of female climate scientists (<https://www.scientificamerican.com/article/as-climate-scientists-speak-out-sexist-attacks-are-on-the-rise/>) serve as a sobering example of direct relevance. We therefore need *considerable* work on this important subject to carefully define best practices for social media use in the present context.

In the meantime, individuals and organizations involved in the Search might:

- familiarize themselves with scholarship describing the problems, such as [96], and the reports and articles cited above
- look at the report from the UN Special Rapporteur on Violence against Women, focusing on online violence worldwide (<https://undocs.org/A/HRC/38/47>)
- consider the resources that UN report recommends: <https://www.accessnow.org/help/> and the Association for Progressive Communications, e.g. the “Due Diligence” project. www.apc.org/sites/default/files/DueDiligenceAndAccountabilityForOnlineVAW.pdf
- consider their own online safety as recommended in [96]. Also, 500 Women Scientists <https://500womenscientists.org/endharassment> recommends The Empowering Internet Safety Guide for Women www.vpnmentor.com/blog/the-empowering-internet-safety-guide-for-women/ and RAINN: <https://www.rainn.org/safe-media>
- consider best practices as employers and supervisors. For example, the Crash Override Network provides concise, useful explanations and guidelines for employers: www.crashoverridenetwork.com/employers.html

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